

LISTEN: Layered Interpretable Speech-to-Text Evidence Network

Team No.: 14

Team Members:

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Explainable Multi-Hop Question Answering over Spoken Audio

Domain

- Speech Language Processing (SLP)
- Deep Learning (DL)
- Explainable Artificial Intelligence (XAI)

Problem Identification

Current spoken question-answering systems can retrieve answers from audio transcripts but struggle to answer questions that require combining evidence distributed across multiple audio segments while providing transparent and verifiable explanations for their reasoning.

Problem Description

Large collections of spoken content such as meetings, lectures, podcasts, interviews, and conference recordings are becoming increasingly common in both academia and industry. Although Automatic Speech Recognition (ASR) systems such as Whisper and Wav2Vec2 can generate highly accurate transcripts, retrieving complex information from these transcripts remains challenging.

Existing spoken question-answering systems generally retrieve one or a few relevant transcript segments and generate an answer without explicitly reasoning across multiple pieces of evidence. Consequently, they perform poorly on questions whose answers require combining information scattered across different timestamps or spoken by different speakers. Furthermore, these systems usually provide only textual responses without explaining which transcript segments contributed to the answer or whether the identified evidence was genuinely necessary. This lack of transparency reduces user trust in critical domains such as healthcare, legal proceedings, business meetings, and education.

The proposed LISTEN framework addresses this limitation by:

1. Converting spoken audio into structured evidence units.
2. Constructing an evidence graph over relevant transcript segments.
3. Performing explicit multi-hop reasoning using a trainable Graph Attention Network (GAT).
4. Generating interpretable answers supported by visual evidence graphs and interactive counterfactual validation.

Relevance to Real-World Scenarios

The ability to accurately answer questions from long spoken recordings has significant applications across many industries, including:

- **Corporate:** Searching meeting recordings for important decisions and action items.

- **Education:** Retrieving detailed explanations from university lecture recordings.
- **Legal:** Finding specific evidence or testimonies within legal hearings.
- **Media:** Summarizing and querying podcast archives.
- **Customer Support:** Identifying trends and evidence in customer support conversations.
- **Healthcare:** Analysis of consultation recordings for medical history verification.
- **Compliance:** Auditing spoken interactions for regulatory adherence.

Providing verifiable evidence alongside every answer increases user trust and makes AI-assisted decision-making more reliable.

Objectives

- To develop a deep learning-based multi-hop spoken question answering framework that leverages Wav2Vec2 for robust speech representation learning and Graph Attention Networks (GAT) for reasoning across multiple semantically related audio segments to accurately answer complex spoken queries.
- To design an explainable evidence retrieval mechanism that identifies and highlights the most relevant speech segments contributing to the generated answer, while incorporating counterfactual reasoning to provide transparent and trustworthy decision explanations.
- To evaluate the proposed LISTEN framework on a synthesized multi-hop spoken question answering dataset using metrics for answer accuracy, retrieval effectiveness, reasoning quality, and explainability, demonstrating its effectiveness over conventional single-hop spoken QA approaches.

Justification

Justification for Objective 1

Current spoken question-answering systems generally treat transcript segments independently and rely primarily on transformer attention or LLM prompting. Such

approaches often fail when answering questions requiring information distributed across multiple parts of long recordings. A Graph Attention Network explicitly models relationships among transcript segments, allowing information to propagate through shared entities and semantic connections. This enables genuine multi-hop reasoning instead of relying solely on implicit transformer attention.

Justification for Objective 2

Most existing explainable AI techniques highlight influential words or generate textual rationales but cannot demonstrate whether specific evidence was actually necessary for producing an answer. By combining SHAP explanations with evidence graph visualization and counterfactual node-removal validation, the proposed system allows users to verify the importance of each evidence segment. This improves transparency, interpretability, and user confidence.

Methodology

1. Speech Language Processing (SLP)

The system first accepts an audio or video recording, with a pipeline consisting of:

- Audio extraction using FFmpeg.
- Automatic Speech Recognition (ASR) using Whisper or Wav2Vec2.
- Timestamp generation for precise referencing.
- Named Entity Recognition (NER) and Semantic parsing.
- **Output Structure:** { segment_id, start_time, end_time, text, entities }

2. Deep Learning Layer

The deep learning module performs reasoning over the structured transcript segments:

- Step 1 (Bi-Encoder Retrieval): Retrieve the Top-K transcript segments relevant to the user's question.

- Step 2 (Evidence Graph Construction): Nodes (Transcript segments); Edges (Shared entities, Shared topics, Semantic similarity).
- Step 3 (Graph Attention Network): Learns explicit multi-hop relationships between transcript segments through message passing.
- Step 4 (Answer Generation): The highest weighted evidence nodes are supplied to an LLM to generate the final answer.

3. Explainable AI (XAI)

- SHAP Explanation: Visualizes which words and transcript segments contributed most during the initial retrieval phase.
- Evidence Graph Visualization: Displays graph attention weights learned by the GAT to show how the model connected different pieces of evidence.
- Counterfactual Validation: Allows users to manually remove an evidence node; the system then reconstructs the graph and re-runs inference to test the necessity of that specific segment.

Team Responsibilities

Role	Responsibility	Assigned To
Speech Processing	Audio/Video upload, FFmpeg extraction, ASR transcription, NER/Semantic parsing.	Tanmoy Saha (23BAI0043)
Deep Learning	Bi-Encoder retrieval, Evidence graph construction, GAT training, LLM answer generation.	Vaibhav Jain (23BAI0033)

Role	Responsibility	Assigned To
Explainability	SHAP explanation, Counterfactual validation, Frontend interface.	Sanjeev Karnatapu (23BAI0003)

Analysis of Existing Approaches

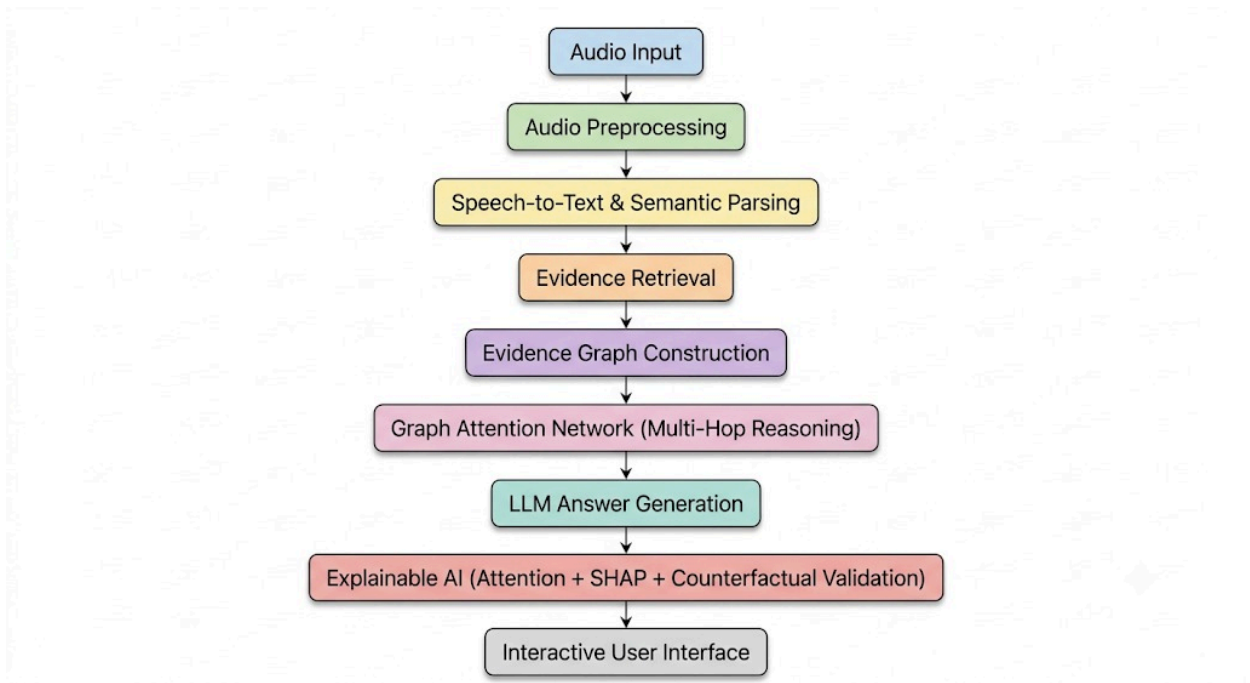
- Strengths: High ASR accuracy from modern models, strong retrieval performance in RAG systems, fluent LLM responses, and established graph-based reasoning (HGN).
- Limitations: Most systems are primarily text-focused, limited to single-hop reasoning, lack support for timestamped audio integration, and offer limited interactive explainability.
- Novel Observations: Integrating speech processing, graph neural reasoning, and interactive counterfactual explainability into a unified framework represents a promising research direction that bridges the gap between raw audio and verifiable insights.

Technical Specifications

Category	Specification
Deep Learning Techniques	Whisper, Wav2Vec2, Bi-Encoder Retrieval, Graph Attention Network (GAT), Transformer Encoder, LLM, SHAP
Standard Datasets	HotpotQA (HGN reproduction), AMI Meeting Corpus (meeting audio), QMSum (query-answering evaluation)

Category	Specification
Base Paper Reference	Hierarchical Graph Network for Multi-hop Question Answering (Microsoft Dynamics 365 AI Research, EMNLP 2020)
Implementation Strategy	Reproduce HGN on HotpotQA, then extend to spoken audio using timestamped segments.

Flowchart



S.No	Title of the Paper	Author(s)	Journal / Conference	Problem Addressed	Methodology	Strengths	Limitations	Research Gap
1	Hierarchical Graph Network for Multi-hop Question Answering	Yuwei Fang, Siqi Sun, Zhe Gan, Rohit Pillai, Shuhang Wang, Jingjing Liu	Proceedings of EMNLP 2020	Multi-hop question answering over multiple Wikipedia documents requiring reasoning across paragraphs, sentences and entities.	Constructs a hierarchical graph with Question-Paragraph-Sentence-Entity nodes initialized using RoBERTa + BiLSTM and performs message passing using a Graph Attention Network (GAT) . Jointly predicts supporting paragraphs, supporting facts, entities and answer spans.	Explicit graph-based reasoning, strong supporting-fact prediction, state-of-the-art performance on HotpotQA, interpretable graph reasoning.	Limited to text-only Wikipedia documents, extractive QA, relies on hyperlinks and structured text, no spoken audio or explainability beyond attention weights.	LISTEN extends HGN to spoken audio , replaces text passages with timestamped transcript segments, uses a segment-level evidence graph, integrates SHAP explanations and introduces counterfactual node-removal validation. (ACL Anthology)
2	Explainable Multi-Hop Question Answering: A Rationale-Based Approach	Kyubeen Han, Youngjin Jang, Harksoo Kim	Big Data and Cognitive Computing (MDPI), 2025	Improve explainability in multi-hop QA without requiring manually annotated rationale labels.	Uses ELECTRA encoder with a Pointer Network to iteratively select supporting rationale sentences while enforcing semantic consistency between selected evidence and predicted answers.	Generates faithful explanations without rationale annotations, lightweight architecture, strong rationale selection performance.	Works only on textual documents, no speech processing, no graph neural reasoning, no interactive explanation.	LISTEN applies explainability to spoken audio , combines GAT reasoning with SHAP explanations and interactive counterfactual validation over timestamped evidence. (MDPI)
3	Thoughtful and Cautious Reasoning: A Fine-Tuned Knowledge Graph-Based Multi-Hop Question Answering Framework	Yinghao Zheng, Ling Lu, Yang Hu, Yinong Chen, Aijuan Wang	Engineering Applications of Artificial Intelligence (Elsevier), 2025	Multi-hop reasoning over knowledge graphs while reducing semantic ambiguity and noisy reasoning.	Proposes TCR-KGQA using a novel question encoder, knowledge graph embeddings and a residual-GRU instruction update framework for iterative reasoning.	Rich semantic understanding, effective noise suppression and accurate graph reasoning.	Requires structured knowledge graphs, cannot process spoken audio or automatically construct evidence graphs from transcripts.	LISTEN automatically builds evidence graphs from Whisper/Wav2Vec2 transcripts and performs GAT-based reasoning with explainable graph visualization and counterfactual validation. (ScienceDirect)

4	Spoken Question Answering and Speech Continuation Using Spectrogram-Powered LLM	Eliya Nachmani, Alon Levkovitch, Roy Hirsch, Julian Salazar, Chulayuth Asawaroengc	ICLR 2024	End-to-end spoken question answering directly from speech without relying entirely on text transcripts.	Uses a speech encoder operating on spectrograms connected directly to an LLM for spoken QA and speech continuation.	End-to-end speech understanding without explicit ASR dependency, strong spoken QA capability.	Focuses on single-hop spoken QA, lacks retrieval over long recordings, graph reasoning, evidence attribution and explainability.	LISTEN extends spoken QA by introducing retrieval across long audio archives, graph-based multi-hop reasoning, evidence graphs and counterfactual explanations.
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